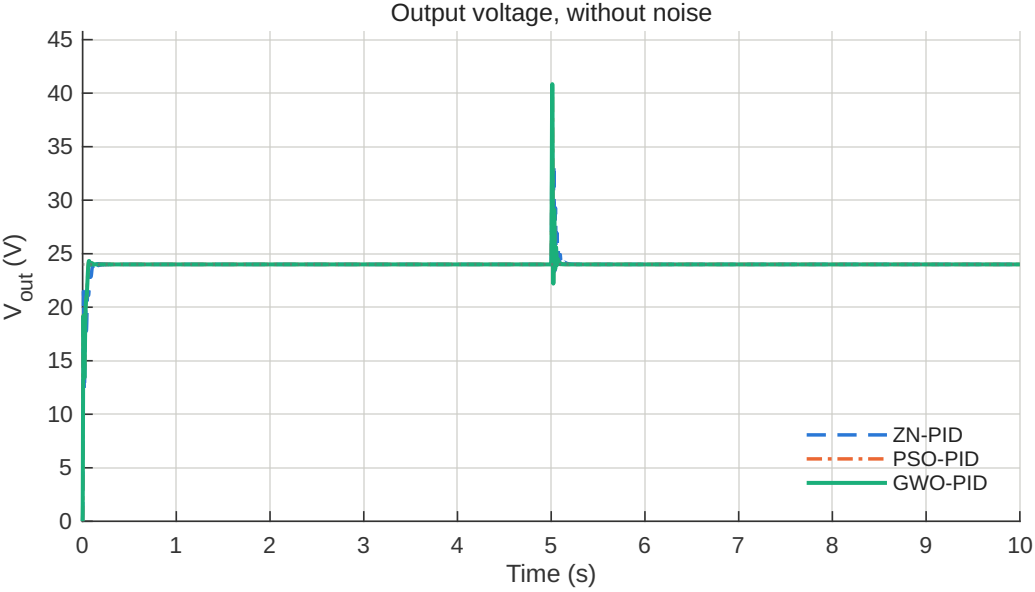

Output voltage, without noise
45
40
35
30
25
20
15
10
5
0
V_{out} (V)
0
1
2
3
4
5
6
7
8
9
10
Time (s)
ZN-PID
PSO-PID
GWO-PID